CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2021 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

Q1 (7.5M) **Explain the functions of the Session Layer in the OSI model.**

Session Layer (Layer 5): Responsible for establishing, maintaining, synchronizing, and terminating communication sessions between applications. Controls dialog (simplex/half-duplex/full-duplex) and inserts synchronization checkpoints into data streams to facilitate restart upon failure.

Q2 (7.5M) **Write short notes on: (a) Throughput & Wavelength, (b) Synchronous Protocols, (c) Routers.**

Notes:

- **(a) Throughput & Wavelength:** Throughput is the actual measurement of data transferred across a network per second. Wavelength (\lambda = v / f) is the physical distance between consecutive signal wave peaks in optical communication.
- **(b) Synchronous Protocols:** Continuous transmission of data blocks synchronized to a shared clock, using sync bytes/flags (e.g. HDLC).
- **(c) Routers:** Network Layer (Layer 3) internetworking devices that inspect destination IP addresses and make packet forwarding decisions based on routing tables.

Q3 (7.5M) **Discuss ISDN, its services, and layer structure.**

ISDN (Integrated Services Digital Network): Transmits voice, data, and video simultaneously over standard telephone copper lines. Channels: B-channel (64 kbps, user data), D-channel (16 or 64 kbps, control signaling). BRI provides 2B+D (144 kbps); PRI provides 23B+D (1.544 Mbps) or 30B+D (2.048 Mbps).

Q4 (7.5M) **Explain Flow Control and Error Control in Computer Networks.**

Mechanisms: Flow control ensures a fast sender does not overwhelm a slow receiver (Sliding Window). Error control detects corrupted frames via CRC and retransmits them via ARQ schemes (Stop-and-Wait, Go-Back-N, Selective Repeat).

Q5 (7.5M) **Explain TCP Packet Format (Header Structure) in detail.**

TCP Header Fields:

- Source Port (16 bits) and Destination Port (16 bits)
- Sequence Number (32 bits) and Acknowledgment Number (32 bits)
- Header Length / Data Offset (4 bits) and Reserved bits (6 bits)
- Control Flags (6 bits: URG, ACK, PSH, RST, SYN, FIN)
- Window Size (16 bits) for flow control
- Checksum (16 bits) and Urgent Pointer (16 bits)

SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

Q6 (15M) **Explain Seven Layers of ISO/OSI Reference Model with their functions.**

Complete OSI Model: Detailed explanation of Physical, Data Link, Network, Transport, Session, Presentation, and Application layers.

SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

Q7 (22.5M) **Explain Routing Algorithms: Shortest Path Routing (Dijkstra) and Distance Vector Routing with Count-to-Infinity problem.**

Routing Algorithms: Dijkstra link-state calculation vs Bellman-Ford distributed vector routing with split horizon and poison reverse solutions.

Q8 (22.5M) **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat with window sizes and frame sequences.**

Sliding Window: Mathematical condition for maximum window sizes: Go-Back-N ($W_s \le 2^m - 1$) and Selective Repeat ($W_s + W_r \le 2^m \implies W_s \le 2^{m-1}$) to prevent ambiguity with duplicate sequence numbers.
